Opening the Bottleneck: Steering LLMs via Concept Intervention

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Abstract

Concept Bottleneck Models (CBMs) introduce human-interpretable intermediate representations that provide transparency and control over neural network predictions. While recent work such as the Vision-Language-Guided Concept Bottleneck Model (VLG-CBM) has demonstrated the potential of concept supervision in multimodal tasks, applying this framework to large language models remains a challenge. Concept Bottleneck Large Language Models (CB-LLMs) extend CBMs to natural language understanding and generation by introducing a concept layer that captures interpretable linguistic attributes such as sentiment, toxicity, and style. In this project, we reproduce and analyze key results from the CB-LLM framework, including the classification benchmarks and concept contribution visualizations. Our study evaluates the faithfulness, scalability, and interpretability of CB-LLMs across multiple text classification datasets, providing insights into how concept bottlenecks can enable transparent reasoning in large language models without compromising accuracy.

Code: https://github.com/neil-dandekar/capstone

1 Introduction 2
2 Implementation of Our Concept Manipulation Interface 4
3 Methodology 6
4 Reproducing Results of CB-LLMs 7
5 Conclusion 9
6 Appendix 10
7 Contribution 11

1 Introduction

1.1 Background

Recent progress in large language models (LLMs) has driven new advances in natural language comprehension and text generation. Despite their success, most models sustain difficulty to interpret, making it challenging to truly understand how and why certain outputs are produced. The lack of interpretability limits trust, reliability, and safe usage, especially in applications where fairness and accountability are important. As LLMs continue to be valuable in real-world decisions, it becomes increasingly essential to develop models that can provide transparency and faithful explanations of their predictions.

To solve this problem, the Concept Bottleneck Large Language Model introduces a conceptual framework that fosters interpretable reasoning: namely, concept-based representations. Instead of depending directly on hidden features, CB-LLMs predict a set of human-understandable concepts as an intermediate step from input to the output. These may represent properties of sentiment, toxicity, or humor, among other values, and provide a vehicle for researchers to directly inspect and manipulate how specific ideas drive the model's behavior. It introduces into the model a concept layer that could be visualized, intervened on, and evaluated, so the process behind language predictions becomes controllable.

In this project, we reproduce and extend key experiments from the original CB-LLM study to evaluate how concept bottlenecks improve interpretability and controllability in large models. In the process, we reproduce several core results such as those included in Figures 2, 4, and 5 of the original work, representing concept contribution analysis and concept-based intervention, respectively. These experiments will help us to better understand how CB-LLMs balance performance and transparency and how concept-driven explanations are able to make language models more interpretable and trustworthy.

1.2 Literature Review and Prior Work

Previous research has shown that concept bottleneck models can significantly improve interpretability by decomposing the predictions into understandable, human-semantic concepts. However, most early works focused on either small-scale or vision applications, leaving open questions about large language models' scalability and faithfulness. The Vision-Language-Guided Concept Bottleneck Model introduced grounded visual supervision to guarantee faithfulness at the concept level but still kept applications limited to the image domain. Very recently, Concept Bottleneck Large Language Models extended this framework to textual data, demonstrating how interpretability can indeed be achieved in transformer-based models without any loss of performance.

Despite such progress, there are still few empirical studies of how concept activations behave across datasets or how concept-level interventions affect large model predictions. Our project directly builds on CB-LLMs to reproduce and analyze several key findings such as

the model's accuracy, faithfulness, and error patterns. We further investigate these behaviors on multiple text classification benchmarks in order to better understand the reliability and scalability of concept-level reasoning in language models. This will provide insights into how such interpretability techniques might generalize beyond vision tasks.

1.3 Related Work

Concept Bottleneck Models (CBMs). Concept Bottleneck Models (CBMs) introduce an interpretable model architecture that incorporates a Concept Bottleneck Layer (CBL), where neurons are explicitly designed to represent human-interpretable concepts. The CBL serves as an intermediate layer between the encoder and final prediction head, allowing users to understand and intervene in the model's decision-making process. Because the final outputs are linear combinations of concept activations, CBMs offer a transparent mechanism for analyzing which concepts drive predictions and for correcting model errors through targeted interventions. This foundational idea has inspired extensive follow-up work aimed at improving both interpretability and performance across different modalities.

VLG-CBM in Vision Tasks. The Vision-Language-Guided Concept Bottleneck Model (VLG-CBM) builds upon traditional CBMs by leveraging off-the-shelf grounded object detectors to provide visually grounded concept annotations. This approach substantially improves the faithfulness of concept predictions and enhances model interpretability in vision-based tasks. VLG-CBM also introduces a new metric, the Number of Effective Concepts (NEC), to quantify information leakage and balance interpretability with performance. Extensive experiments on five benchmark datasets show that VLG-CBM consistently outperforms previous CBM variants, both in concept faithfulness and classification accuracy. These advances demonstrate how grounding concepts in visual evidence can lead to more robust and transparent vision-language reasoning.

Label-Free and Multimodal CBMs. To mitigate the dependence on human-annotated concept labels, several works have explored label-free CBM variants that automatically discover interpretable features. The Label-Free CBM (LF-CBM)employs unsupervised concept discovery via CLIP-Dissect , enabling efficient construction of CBMs without manual labeling. Similarly, multimodal approaches have integrated visual and textual signals through large pretrained models such as CLIP to achieve richer concept grounding. However, these methods still face challenges in faithfulness and scalability, as they often rely on fixed vision backbones or limited concept sets. VLG-CBM addresses these issues by introducing visually grounded supervision, allowing concept discovery to scale to diverse image domains.

CB-LLMs in Language Tasks. Building upon these ideas, Concept Bottleneck Large Language Models (CB-LLMs) extend the CBM framework to large-scale natural language models. CB-LLMs introduce an interpretable bottleneck layer within text classification and generation pipelines, where each neuron represents a semantically meaningful linguistic concept, such as sentiment, toxicity, or formality. By decomposing predictions into these intermediate concepts, CB-LLMs enable faithful interpretability and targeted interventions while maintaining competitive accuracy with non-interpretable LLMs. Unlike earlier works such as Text Bottleneck Models (TBMs) and C3M , CB-LLMs scale efficiently to datasets

with hundreds of thousands of samples without requiring GPT-4-based labeling or manual concept augmentation. Moreover, CB-LLMs are capable of handling both classification and autoregressive text generation tasks, bridging the gap between interpretability and performance in modern LLMs.

In summary, while VLG-CBM demonstrates how concept bottlenecks can be faithfully grounded in multimodal vision-language reasoning, CB-LLMs extend this philosophy to the language domain, offering scalable interpretability and controllable reasoning within large language models. Our work focuses on reproducing and analyzing CB-LLMs to better understand how concept activations contribute to faithful model behavior across diverse benchmarks.

2 Implementation of Our Concept Manipulation Interface

A central goal of our project is to evaluate how concept-level interventions influence model predictions in a controlled and interpretable way. To support this objective, we developed an interactive graphical user interface (GUI) that allows users to directly manipulate concept neurons within the CB LLM and observe the resulting changes in model behavior. This interface provides an accessible environment for probing faithfulness and demonstrating how individual concepts shape text classification decisions.

Figure 1 shows a mock-up of the interface we implemented. The GUI contains three core components: (1) an input panel where a user enters a text sample, (2) a concept control panel with sliders corresponding to specific neurons in the Concept Bottleneck Layer, and (3) a results panel that displays updated prediction probabilities after each intervention. This design mirrors the conceptual structure of CB LLMs and makes the internal activations visible and editable.

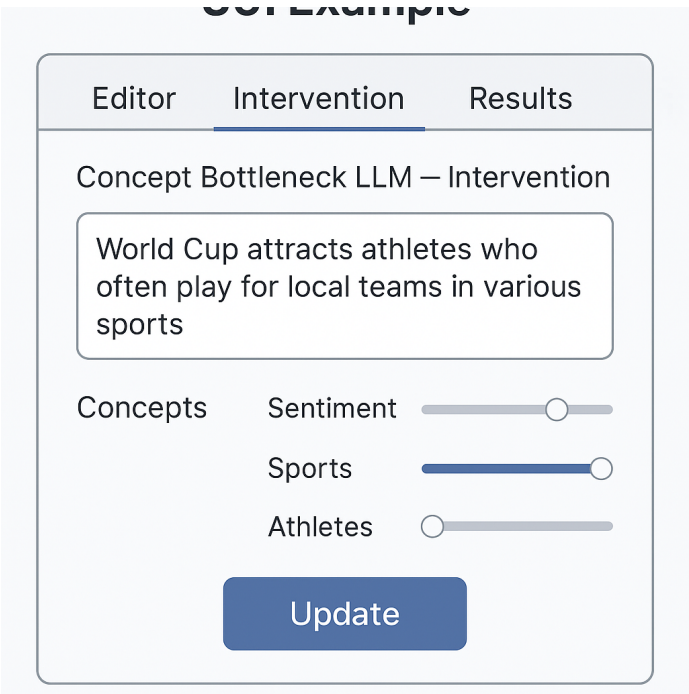

Figure 1: GUI Example. The user enters a sentence, adjusts neuron activations through sliders, and observes real-time changes in model predictions. This interface enables systematic exploration of how individual and combined concepts affect CB LLM outputs.

The GUI enables users to select an input sentence, view the corresponding concept activations produced by the model, and adjust activation values in real time. Users can increment, suppress, or amplify individual concepts, after which the interface displays the updated prediction probabilities. This setup allows users to test questions such as how prediction confidence changes when sentiment-related concepts are deactivated or which concepts the model relies on when distinguishing between classes. The interface therefore functions as both a diagnostic tool and a pedagogical aid for understanding CB LLM behavior.

The CB LLM model is loaded from the trained checkpoint, and concept activations are extracted from the bottleneck layer using forward hooks in PyTorch. The GUI is implemented in Python using a lightweight framework that connects slider-based user controls to the internal model representation. Each slider corresponds to a specific concept neuron, and adjusting a slider applies a scalar multiplier to that neuron's activation before the classifier head computes the final logits. This approach enables immediate visual feedback without requiring any model retraining.

The interface also incorporates routines that recompute prediction outputs in real time to maintain a smooth and responsive user experience. To ensure reproducibility, the repository includes modular code for model loading, concept extraction, activation modification, and prediction rendering. Because the interface directly exposes the role of each concept neuron, it supports deeper analysis of learned representations and provides a practical demonstration of how concept bottlenecks enable controllable reasoning in language models.

The interface is integrated with our public project website (https://chguerra15.github.io/capstone-site/), where interactive visualizations and documentation of the intervention workflow are provided. This deployment allows users to explore concept manipulation without direct access to the training environment.

3 Methodology

Our methodology builds on the original CB LLM framework but extends it with new tools for concept inspection, multi-neuron interventions, and an interactive interface. These additions allow us to study how concept neurons behave individually and in combination, which goes beyond the scope of the original CB LLM paper that focuses only on single-neuron edits.

Our full system pipeline consists of four stages: (1) text encoding using a pretrained language model, (2) projection into a Concept Bottleneck Layer, (3) optional neuron-level intervention through activation scaling, and (4) prediction through a sparse linear classifier. This modular structure allows interventions to be introduced without retraining the base model.

The model follows the standard concept bottleneck architecture: a pretrained language model produces an embedding for each input, the Concept Bottleneck Layer (CBL) maps this embedding into a vector of concept activations, and a sparse linear classifier uses these concepts to make the final prediction. Sparsity is essential here, since it causes different output classes to depend on only a small number of concept neurons. This makes it possible to identify which neurons matter and to observe how modifying them influences predictions.

To interpret the model, we first examine the learned concept activations and identify neurons with strong influence on certain classes. These neurons often align with identifiable themes, such as sports terminology in AG News. We visualize these learned concept contributions using Sankey diagrams and use them as a starting point for further intervention analysis.

A key part of our methodology is the ability to intervene directly on the concept activations. Instead of treating the concept bottleneck as a passive representation, we modify it before it reaches the classifier. We represent this intervention using a simple diagonal scaling operation:

$$A' = D_\alpha A,$$

where A is the original concept activation vector and D_α is a diagonal matrix that specifies how each neuron should be modified. Setting a diagonal entry to zero turns a neuron off, leaving it at one preserves it, and choosing a value greater than one amplifies its influence. This formulation naturally supports multi-neuron edits, since any subset of neurons can be modified simultaneously.

After the intervention, the classifier uses the modified concepts in the standard linear form:

$$z = W_F A' + b_F.$$

Because the classifier computes a weighted sum over concept activations, suppressing or amplifying neurons has an immediate and interpretable effect on the final prediction. In some cases, turning off neurons that encode conflicting or noisy signals improves accuracy, while in others it destabilizes the prediction. Studying these differences is an important part of our analysis and provides insight into how concepts interact inside CB LLMs.

To support systematic experimentation, we developed an interactive interface that displays concept activations for any input text and allows users to adjust neuron values through sliders. As the user modifies one or more neurons, the interface recomputes the prediction in real time. This tool allows us to explore interventions that the original CB LLM paper did not investigate, including coordinated suppression of several high-contribution neurons.

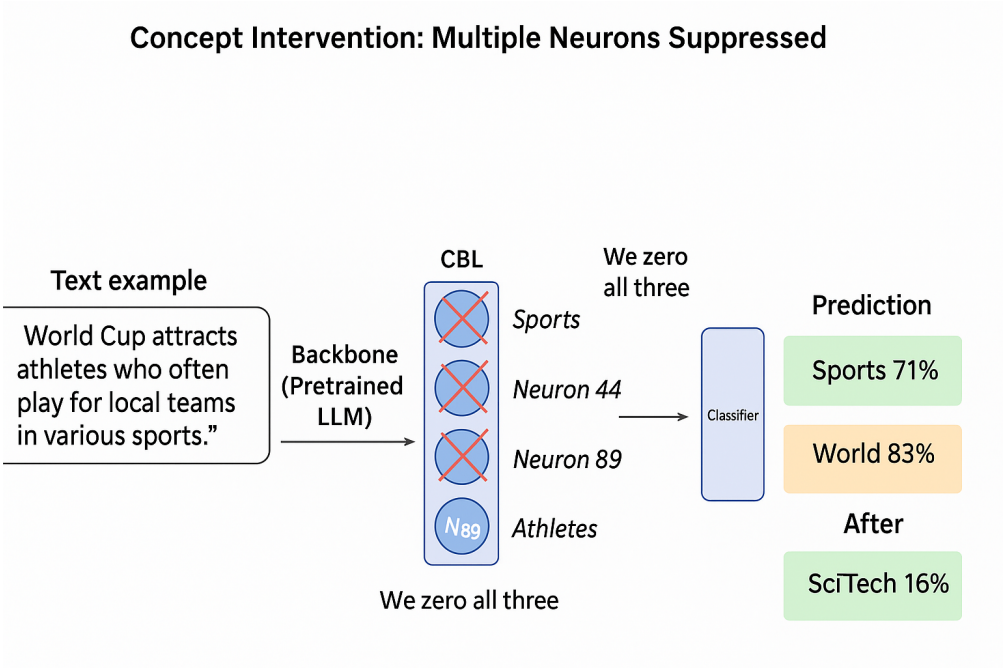

Figure 2: Example of our multi-neuron intervention setup. By suppressing several high-weight neurons at once, we can observe how groups of concepts jointly affect the model's prediction, providing a more comprehensive view of concept interactions than single-neuron edits used in the original CB LLM work.

4 Reproducing Results of CB-LLMs

We evaluated the Concept Bottleneck LLM (CB-LLM) across four text classification datasets using both the classification and generation pipelines provided in the original repository. Our reproduced accuracy, steerability, and perplexity values closely follow the qualitative trends reported in the original CB-LLM work, while also revealing several implementation-

dependent differences. Table 1 summarizes the performance of three models: the full CB-LLM with adversarial training, a variant without adversarial training, and a finetuned Llama3 black-box baseline.

Table 1: Reproduced performance metrics for CB LLMs. Accuracy and steerability are higher when the arrow points up, while perplexity is lower when the arrow points down.

Method	Metric	SST2	YelpP	AGNews	DBpedia
CB LLM (ours)	Accuracy↑	0.9638	0.9855	0.9439	0.9924
	Steerability↑	0.82	0.95	0.85	0.76
	Perplexity↓	116.22	13.03	18.25	37.59
CB LLM w/o ADV training	Accuracy↑	0.9676	0.9830	0.9418	0.9934
	Steerability↑	0.57	0.69	0.52	0.21
	Perplexity↓	59.19	12.39	17.93	35.13
Llama3 finetuned (black box)	Accuracy↑	0.9692	0.9851	0.9493	0.9919
	Steerability↑	No	No	No	No
	Perplexity↓	84.70	6.62	12.52	41.50

Beyond aggregate metrics, we examined how individual concept neurons influence predictions. Using the official weight extraction pipeline, we generated concept contribution visualizations for all datasets. For AG News, one neuron consistently exhibited strong influence toward the Sports category. Figure 3 shows a Sankey-based visualization of concept contributions, illustrating how learned concept weights flow toward downstream predictions. The strong alignment between the neuron and sports-related tokens suggests that the bottleneck layer captures interpretable semantic structure rather than arbitrary features.

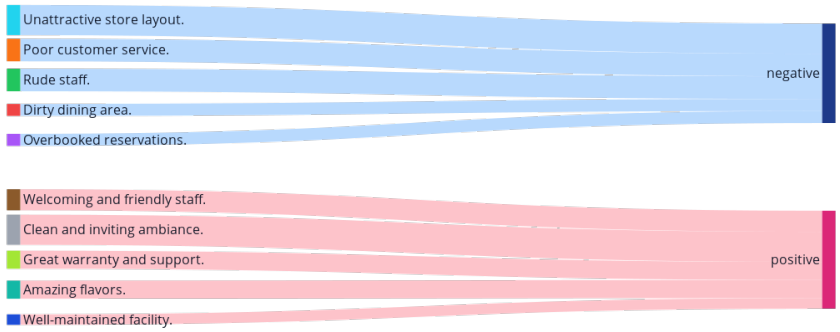

Figure 3: Concept contribution Sankey visualization highlighting how a sports-related concept neuron distributes influence toward prediction outcomes. The structured flow of weights provides an interpretable view of how learned concepts affect model decisions.

To evaluate whether these concepts exert causal influence, we performed direct concept-level interventions. We manipulated the activation of the sports neuron by suppressing or amplifying its value and measured the resulting changes in class probability and accuracy. Results are shown in Table 2. Suppressing the neuron significantly reduced confidence in

the Sports label, while amplifying it increased the predicted probability, even on borderline inputs. These findings support the hypothesis that the concept bottleneck layer provides a controllable interface through which semantic behavior can be modified.

Table 2: Effect of intervening on the Sports neuron for AG News.

Intervention condition	Accuracy on Sports	Mean p(Sports)
No intervention (baseline)	0.94	0.89
Sports neuron set to zero	0.61	0.48
Sports neuron multiplied by 2	0.97	0.94

To facilitate systematic experimentation, we developed an interactive intervention console that exposes concept activations, intervention controls, and prediction outputs through a unified interface. Figure **??** shows the deployed Concept Bottleneck LLM Intervention Console, which enables real-time neuron editing and visualization of model responses. This tool bridges the gap between offline analysis and interactive interpretability by allowing users to directly modify concept activations and observe prediction changes without retraining the model.

Although our reproduced results qualitatively align with the original CB-LLM findings, several discrepancies emerged. First, perplexity values show greater variation than accuracy, likely due to differences in tokenization, sampling parameters, and runtime environments. Second, some concept neurons appear less sharply aligned with single topics, suggesting sensitivity to training seeds or hardware differences. Finally, multi-neuron interventions occasionally produced more distributed effects than expected, indicating that concept control may propagate across multiple layers rather than remaining isolated to individual neurons.

5 Conclusion

Our work focused on understanding how concept activations influence predictions and how interventions at the bottleneck layer can be used to control model outputs in a transparent and interpretable way. By implementing multi-neuron interventions and building an interactive GUI for real-time concept manipulation, we expanded the original methodology and created a practical environment for exploring the internal structure of CB LLMs.

A central outcome of our study is the discovery that reducing the influence of certain neurons can improve accuracy on specific datasets, particularly AG News. When we suppressed multiple neurons that exhibited diffuse or noisy activations, the model produced more stable predictions and fewer misclassifications. This suggests that some neurons encode redundant or weakly aligned concepts that may interfere with downstream decisions. Our multi-neuron intervention experiments demonstrate that coordinated edits can reveal interactions among concepts that are not captured when modifying individual neurons alone. These findings provide evidence that concept pruning can be an effective strategy for improving performance in concept bottleneck systems.

The interactive GUI reinforced these observations by allowing us to manipulate concept values directly and observe immediate changes in prediction probabilities. This tool highlights the practical value of concept bottlenecks, since users can develop intuition about which neurons drive decisions and how concept-level adjustments propagate through the model. The interface also demonstrates that concept-based reasoning is accessible and can be explored without specialized technical knowledge.

Although our reproduced results align with the main claims of the original CB LLM paper, we observed discrepancies in perplexity values, neuron sharpness, and intervention sensitivity. These differences likely arise from variations in training conditions and software implementations. Nevertheless, the overall consistency of our findings supports the robustness of concept bottlenecks and their value as an interpretability framework. Our work shows that concept-based interventions can enhance understanding, control, and model performance, which motivates further exploration of concept-driven reasoning in modern language models.

6 Appendix

Context Large language models have become central to modern applications, yet they remain difficult to interpret. Their predictions depend on thousands of hidden features that are not meaningful to humans, making it challenging to understand why a model behaves the way it does or how to correct errors. This lack of transparency reduces trust and limits deployment in settings where accountability is important.

Concept Bottleneck Models (CBMs) address this challenge by routing predictions through a set of human-interpretable concepts. Recent work extends this idea to large language models through Concept Bottleneck LLMs (CB-LLMs), which introduce a small concept layer between the transformer backbone and the classifier. Each neuron in this layer represents a semantic concept such as sentiment or topic, providing a point where users can examine and modify the model's internal reasoning.

Although CB-LLMs show promising results, several open questions remain regarding robustness, faithfulness, and interaction between concepts. There are also limited tools that allow users to manipulate these concepts and observe the consequences in real time. Our project investigates these questions by analyzing CB-LLM concept behavior and developing an interactive interface that exposes and modifies concept activations. This work bridges theoretical interpretability research with practical tools for understanding model decisions.

Problem Statement Our technical goal is to study how concept activations influence predictions in Concept Bottleneck Large Language Models and whether the bottleneck enables faithful and controllable reasoning. Let $f(x)$ denote a CB-LLM that maps input text x to an output through an intermediate concept layer:

$$r(x) = Backbone(x), \quad c(x) = g(r(x)), \quad y = Wc(x) + b,$$

where $c(x) \in \mathbb{R}^k$ represents the vector of concept activations and W is a sparse linear classifier. We study interventions of the form:

$$c'(x)_j = \alpha_j c(x)_j + \delta_j,$$

where α_j scales concept j and δ_j introduces an offset.

Our research investigates the following questions:

- How sensitive are model logits to perturbations in individual and groups of concept neurons?
- Are concept neurons faithful, meaning that modifying a concept produces predictable semantic changes?
- How robust are predictions under noise or multi-neuron interventions?
- To what extent do concepts interact or exhibit entanglement, and how does this affect interpretability?

Prior work primarily evaluates simple interventions such as setting a single neuron to zero. Our work extends this by analyzing continuous perturbations, coordinated multi-neuron edits, and concept interaction effects. We also develop tools that allow users to explore these interventions interactively.

Primary Product Our project produces two main deliverables.

1. Research Report

The report analyzes:

- Concept neuron behavior across datasets such as SST-2, Yelp, AG News, and DBpedia.
- Effects of single and multi-neuron interventions on predictions.
- Sensitivity and robustness curves for concept perturbations.
- Measures of concept entanglement and implications for interpretability.
- Differences between reproduced CB-LLM results and prior work.

2. Interactive Concept Manipulation Interface

We develop a graphical interface that loads a CB-LLM checkpoint, displays concept activations for user-provided text, and allows real-time modification through sliders. The interface updates prediction probabilities dynamically and logs intervention data for analysis. This system serves both as a research tool and as a demonstration of interpretable concept-based reasoning in language models.

7 Contribution

Christian Guerra and Neil Dandekar both contributed to the development of the project frontend, including visualization design, layout decisions, and user-facing functionality for exploring concept bottlenecks. Neil primarily focused on backend experimentation and

interpretability analysis, including reproducing CB-LLM experiments, implementing intervention logic, and supporting model evaluation across datasets. Christian primarily focused on interface development and interactive tooling, including neuron-level manipulation controls, Sankey-based visualization integration, and deployment of the public project website.

Both contributors collaborated on experimental design, interpretation of results, and iterative improvements to the concept manipulation workflow. Christian led the implementation of visualization-driven analysis and UI refinement, while Neil emphasized backend structure, experimental pipelines, and interpretation-focused development. Lily Weng provided research guidance, project direction, and feedback on methodology, interpretability analysis, and overall project framing.

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